Topic Modelling

Discovering Themes in Text

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What Is Topic Modelling?

Key idea

An **unsupervised** method for discovering **latent themes** (topics) in a collection of documents — no labels or pre-defined categories required.

Analogy: imagine a reader sorting a large pile of documents into thematic groups *without being told what the themes are*. Topic modelling automates this process.

- Documents are **mixtures** of topics — a speech can be partly about health, partly about the economy
- The algorithm discovers topics by identifying **word co-occurrence patterns**
- Widely used in political science, sociology, media studies, and digital humanities

LDA Intuition: The Generative Story

Latent Dirichlet Allocation (Blei, Ng & Jordan, 2003) imagines each document was "generated" by:

1. Choose a mix of topics → 2. For each word, pick a topic → 3. Pick a word from that topic

Example "generated" document:

health **NHS** **tax** **care** **budget** **patients** **growth**

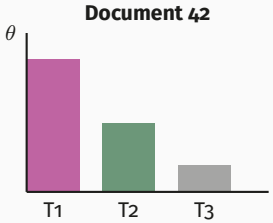
■ Health topic ■ Economy topic

LDA *reverses* this process: given the observed words, it infers the hidden topic structure.

LDA: Two Key Distributions

(a) Document → Topics

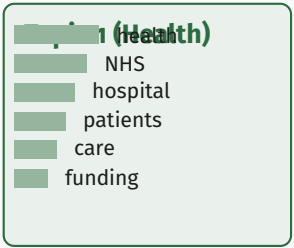
Each document has a distribution over topics.



Document 42 is mostly about Topic 1, somewhat about Topic 2, and a little about Topic 3.

(b) Topic → Words

Each topic has a distribution over words.



The highest-probability words characterise the topic's theme.

Example: Topics from Parliamentary Debates

LDA with $K = 4$ topics fitted to UK House of Commons debates. Top 10 words per topic:

Topic	Top 10 Words
1 — Health	health, NHS, hospital, patients, care, funding, treatment, service, staff, mental
2 — Education	education, school, university, students, teachers, learning, skills, children, fund
3 — Economy	economy, growth, tax, business, employment, trade, investment, inflation, bud
4 — Security	security, defence, military, police, terrorism, intelligence, border, crime, forces,

Observation

The algorithm has no knowledge of politics — it discovers these groupings purely from **word co-occurrence patterns**. Labels (Health, Education, etc.) are assigned by the researcher after inspection.

Choosing K: The Number of Topics

Why it matters

K is set *by the researcher* before fitting the model. Too few topics conflate distinct themes; too many produce fragmented, uninterpretable topics.

Approaches to selecting K:

- **Coherence scores** — automated measure of how semantically related top words are within each topic
- **Held-out likelihood** — how well the model predicts unseen documents
- **Human judgement** — do the topics make substantive sense to domain experts?

Practical advice

Try multiple values of K (e.g., 5, 10, 15, 20). Inspect and compare results. There is **no single "correct"** K — the best choice depends on your research question.

Interpreting and Validating Topics

How to interpret topics:

- Read the **top words** for each topic — do they form a coherent theme?
- Examine **high-loading documents** — read texts most strongly associated with each topic
- Assign **labels** carefully — resist the temptation to over-interpret a word list

External validation:

- Compare topic assignments with **known metadata** (e.g., committee, policy area)
- Check topic prevalence over **time** — do trends match known events?

Common pitfalls

Junk topics (stopwords, boilerplate) · **Over-splitting** (one theme spread across several topics) · **Label bias** (reading meaning into vague word lists)

Structural Topic Models (STM)

STM extends LDA by allowing **document-level metadata** (e.g., party, year, author) to influence topic **prevalence** and **content**.

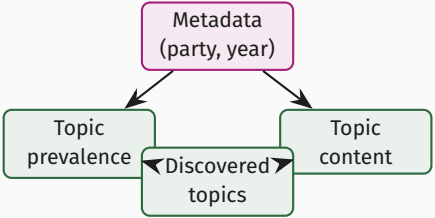
What STM adds:

- **Topic prevalence** can vary with covariates

  e.g., "Health" topic more prevalent in Labour speeches

- **Topic content** (word use) can vary

  e.g., Labour and Conservative use different words when discussing health



Key reference

Roberts, Stewart & Tingley (2014). *"stm: R Package for Structural Topic Models."* See also Roberts et al. (2014), *American Journal of Political Science*.

Strengths and Limitations

Strengths

- **Scalable** — can handle thousands or millions of documents
- **No labelled data required** — fully unsupervised
- **Exploratory power** — discovers themes the researcher may not have anticipated
- **Mixed membership** — documents can belong to multiple topics

Limitations

- **Bag-of-words** — ignores word order and syntax
- **Sensitive to** K — results change with the number of topics
- **Preprocessing matters** — stopwords, stemming, and rare-word thresholds affect output
- **Not deterministic** — different random seeds can yield different solutions

Topic modelling is best used as an **exploratory** tool. Always validate results with close reading and domain knowledge.

Practical 3: Topic Modelling

- Preprocess a corpus of parliamentary debates for topic modelling
- Fit LDA models with different values of K
- Inspect topic–word distributions and document–topic assignments
- Evaluate topic quality using coherence scores
- Visualise topics and their prevalence over time

Open the **Practical 3** notebook in Google Colab
(Python or R — your choice!)